University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 25
Program: B.Sc. in CSE
2nd year 2nd Semester

Course Title: Database Systems | Course Code: CSE 211
Time: 3:00 hours | Credit Hour: 3.00 | Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS] (see note)

a. "Transaction management can be explained by employing ACID properties"-Explain this statement in terms of ACID properties. [10] CLO1/PLO-a

QUESTION 2 [10 MARKS]

a. Construct the decomposed tables applying 3NF in the table below: [5] CLO3/PLO-d

Product ID	Product Name	Supplier ID	Supplier Name	Supplier City
P01	Laptop	S01	HP	New York
P02	Printer	S02	Canon	Chicago
P03	Monitor	S01	HP	New York
P04	Mouse	S03	Logitech	Oregon
P05	Keyboard	S03	Logitech	Oregon

b. Construct the decomposed tables applying 1NF and 2NF in the table below: [5] CLO3/PLO-d

StudentID	StudentName	Subjects	Marks
1	Leonardo	Math, English	85, 78
2	Lilian	Math	92
3	Allen	English, Science	80, 88
4	Alex	Math, Science	75, 82

QUESTION 3 [10 MARKS]

a. The following relational schema is part of a hotel management database: [10] CLO2/PLO

Student (S_ID, S_First_Name, S_Last_Name, S_Department)
Course (C_ID, C_Name, C_Credit)
Enrollment (E_ID, S_ID, C_ID, Grade)
Instructor (I_ID, I_First_Name, I_Last_Name, I_Salary, C_ID)

Construct the *SQL commands (DML)* for the following queries:

i) Find the course name that has the lowest credit value using a subquery.

ii) Display the names of students and the courses they are enrolled in.

iii) Retrieve the last name of the instructor who receives the highest salary.

iv) List the course IDs where the course name starts with 'C' and ends with 's'.

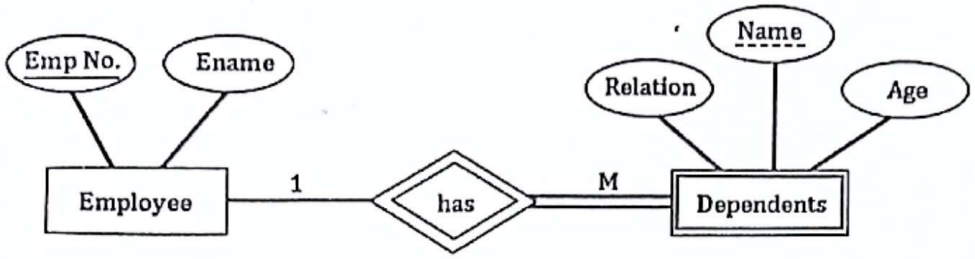
v) Find the instructor IDs who teach students enrolled in 3-credit courses, using a set operation.

QUESTION 4 [10 MARKS]

a. A Movie Production Company manages a collection of movies that are available for streaming on different platforms. Each movie is identified by a unique movie_id and has attributes such as title, release_year, and duration. Every movie is directed by one or more directors, each identified by director_id, with attributes name, experience_years, and an optional contact_number. Each movie features several actors, and each actor (actor_id, name, date_of_birth, nationality) can appear in multiple movies. The company records the roles played by actors in each movie. Movies are released on one or more streaming platforms (platform_id, name, subscription_type), and each platform can stream multiple movies. Every movie belongs to a specific genre (genre_id, genre_name), but a genre can include many movies. Some platforms also host exclusive shows (show_id, title, episodes_count), which are managed separately but may share directors with the movies. [7] CLO4/PLO

Analyze the scenario described above using an appropriate Entity-Relationship Diagram (ERD).

b. Analyze this statement- "A Weak Entity depends on a Strong Entity for its own existence" using the following diagram [3] CLO4/PLO-c



QUESTION 5 [10 MARKS]

a. Explain the responsibilities of any two database users. [5] CLO1/PLO-a

b. Compute the closure of the following set F of functional dependencies for relation schema R = (A, B, C, D, E) [5] CLO3/PLO-d

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Show that the relation $E \rightarrow AD$ holds in the closure of F.

OR

a. "In order to remove the disadvantages of file management system, DBMS is introduced'- Explain how DBMS eliminates the file management system's disadvantage? [5] CLO1/PLO-a

b. Compute the closure of the following set F of functional dependencies for relation schema R = (A, B, C, X, Z) [5] CLO3/PLO-d

$F = \{A \rightarrow B, C \rightarrow X, BX \rightarrow Z\}$

Show that the relation $AC \rightarrow Z$ holds in the closure of F.

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
2nd year 2nd Semester

Course Title: Digital Logic Design | Course Code: CSE 209
Time: 3:00 hours | Credit Hour: 3.00 | Full Marks: 50

There are Five Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

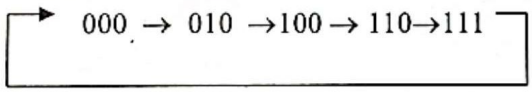
a. Discuss the universality of NOR gate. [3] CLO3/PLO4

b. Implement the following Boolean function with only NOR gate: [3] CLO3/PLO4
$y = AB`C` + BC + D.$

c. A 4-bit binary number is represented as D C B A, where D, C, B and A represent the individual bits with A equal to the LSB. Design a logic circuit that will produce a HIGH output whenever (the binary number is greater than 0101 and less than 1011) or (all inputs are LOW) or (all inputs are HIGH) or (exactly 2 inputs are LOW). [4] CLO3/PLO4

QUESTION 2 [10 MARKS]

a. Design a synchronous counter that will count in this following sequence: [10] CLO4/PLO5



QUESTION 3 [10 MARKS]

a. Briefly describe the operation of IC # 7483(4-bit parallel adder). [2] CLO2/PLO1

b. Draw a 4-bit parallel Adder/Subtractor using IC # 7483 and basic gates if necessary. Briefly describe its operation. [2] CLO2/PLO1

c. Draw a BCD adder using IC # 7483 and NOR gates only. Briefly describe its operation. [6] CLO2/PLO1

QUESTION 4 [10 MARKS]

a. Design the circuit diagram of MOD-13 synchronous up counter using JK flip-flops and briefly describe its operation. [4] CLO5/PLO5

b. Design MOD 100 counter using IC # 74293. [2] CLO5/PLO5

c. Design MOD 6 Ring and Johnson counter using JK flip flop and describe their operation. [4] CLO5/PLO5

QUESTION 5 [10 MARKS]

a. Draw the internal circuit of IC # 74138(Decoder). Briefly describe its operation. [4] CLO3/PLO4

b. Show that IC # 74138(Decoder) can be used as the 1of 8 Demultiplexer. [2] CLO3/PLO4

c. Draw 5 lines to 32 lines decoder using IC# 74138. You can use other logic gates if necessary. [4] CLO3/PLO4

OR

a. Draw the internal circuit of IC # 74151(Multiplexer). Briefly describe its operation. [4] CLO3/PLO4

b. Show how IC # 74151 can be used to generate the logic function Z = AC` + A`B` + BC`. [2] CLO3/PLO4

C. Implement the function F (A, B, C, D) = Σ(0, 2, 3, 4, 8, 7, 9, 11, 13,15) using an IC # 74151 and NOR gate only. [4] CLO3/PLO4

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
2nd year 2nd Semester

Course Title: Engineering Economics
Time: 3:00 hours | Credit Hour: 3.00 | Course Code: ECN 201 | Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. Describe GDP and GNP. [4]

b. Explain different methods of calculating GDP. [6]

QUESTION 2 [10 MARKS]

Illustrate the concept of Marginal Rate of Substitution (MRS) using an indifference curve. [10]

OR

Explain the properties of an indifference curve. [10]

QUESTION 3 [10 MARKS]

Explain different types of market structure. [10]

QUESTION 4 [10 MARKS]

Explain consumer equilibrium under indifference curve analysis. [10]

QUESTION 5 [10 MARKS]

a. Explain inflation. [4]

b. Calculate CPI and inflation rate. [6]

Category	Weight	Year 1	Year 2
Food	0.175	270	270
Housing	0.460	300	330
Transportation	0.193	180	180
Medical Care	0.086	300	330
Other	0.086	250	260

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
2nd year 2nd Semester

Course Title: Data Structures and Algorithms II | Course Code: CSE 207
Time: 3:00 hours | Credit Hour: 3.00 | Full Marks: 50

There are **Five*** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. A toy vending machine dispenses small collectible toys and accepts toy tokens of different values: 1, 5, 10, 20, and 50 tokens. Sometimes, due to stock limitations, the machine cannot use all types of tokens. Currently, the machine has the following tokens in stock: [5] CLO2/ PLO-c

Token Value	Available Tokens
1 token	5
5 tokens	4
10 tokens	3
20 tokens	3
50 tokens	2

A customer redeems a gift card worth 500 tokens and purchases a toy set priced at 371 tokens. The machine now needs to return 129 tokens as change, using the least number of tokens possible without exceeding the available stock.
Design an appropriate algorithm to find the combination of tokens (using no more than the available quantity of each) that adds up to the targeted value and implement that algorithm in the above scenario.

b. A software company is developing a text-comparison tool to detect plagiarism by finding the Longest Common Subsequence (LCS) between two research papers. An intern first implemented the algorithm using simple recursion, but when tested on large texts (strings of length 1000+), the program became extremely slow and often froze. As a senior engineer, analyze why the simple recursive LCS approach is inefficient for large inputs. [2] CLO3/ PLO-b

c. Propose a Dynamic Programming solution to optimize the problem of Question 1.(b). [3] CLO2/ PLO-c

QUESTION 2 [10 MARKS]

a. A telecommunication company, NetLink Pvt. Ltd., plans to lay optical fiber cables to connect its six regional offices located in different cities and it has a head office which A. The goal is to connect all offices using the minimum total cable length while ensuring that every office is connected directly or indirectly to every other office. The following table shows the distances (in kilometers) between the regional offices where laying cable is possible: [6] CLO2/ PLO-c

Table 1: NetLink Pvt. Ltd. Data

From	To	Distance (Km)
A	B	4
A	C	6
A	D	16
B	C	24
B	E	10
C	D	8
C	E	12
D	E	14
D	F	18
E	F	30

Now, design and implement an appropriate algorithm to connect all offices using the minimum total cable length. (constraint: you must have to start your connection from head office).

b. The city of Metropolis is developing software to analyze how its metro stations are connected through underground tunnels. Each station is represented as a node, and each tunnel is represented as a directed link between two stations. The system should be able to explore all possible routes starting from a given station, moving as far as possible along each path before backtracking to explore alternative routes. Develop an algorithm that explores the metro system starting from a given station and records the order in which the stations are visited.. [4] CLO2/ PLO-c

OR

a. The National Power Grid Corporation is planning to connect several power substations across different cities with minimum total wiring cost. Every substation must be connected either directly or indirectly so that electricity can be distributed throughout the network. The company has surveyed the cost of laying electrical cables (in lakhs of rupees) between substations as shown below: [6] CLO2/ PLO-c

Table 2: NPGC Data

Connection	Cost(In Lakhs)
A-B	7
A-D	5
B-C	8
B-D	9
B-E	7
C-E	5
D-E	15
D-F	6
E-F	8

Now, design and implement an appropriate algorithm to connect all substations using the minimum laying electrical cables.(constraint: sorting can only be applied once).

b. A logistics company, TransRoute Pvt. Ltd., manages transportation among several cities. Each city is connected to some others through direct routes, and every route has an associated travel cost (in units of fuel).However, not all cities are directly connected, and the company wants to find the least travel cost between every possible pair of cities, even if it requires passing through intermediate cities. Develop an algorithm that determines the minimum travel cost between all pairs of cities, considering all possible intermediate cities that could reduce the total cost. [4] CLO2/ PLO-c

QUESTION 3 [10 MARKS]

a. Consider the following graph: [5] CLO4/ PLO-c

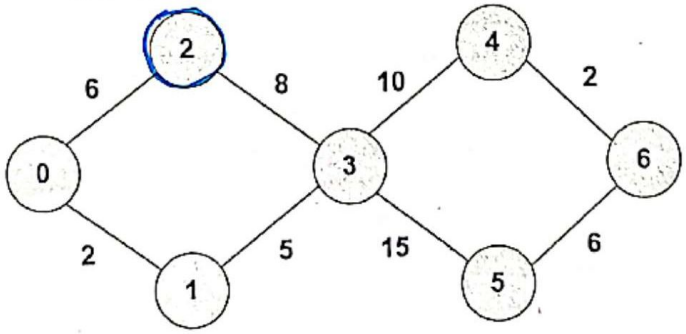


Figure 1. Graph of different connected nodes

Apply Dijhktra's algorithm in the above graph (Figure 1.), find the shortest path and distance from source node 2 to all other remaining nodes.

b. A transportation company, CityTrans Logistics, is analyzing the cost of transporting goods between several cities. Some routes have toll discounts (negative costs) due to special trade agreements. The company wants to find the [5] CLO4/ PLO-c

shortest path from its main hub (City A) to all other cities. The transportation network is represented through the following table:

Table 3: CityTrans Logistics Data

From	To	Cost
A	B	4
A	C	5
B	C	-3
B	D	6
C	D	3
D	B	-8

i. Represent the above network as a directed weighted graph.

ii. Apply the Bellman-Ford algorithm to find the shortest path cost from source vertex A to all other vertices.

iii. Detects whether there is a negative weight cycle in the graph based on Question 3.(b).(iii). If a negative weight cycle exists, explain how it affects the shortest path calculation.

QUESTION 4 [10 MARKS]

a. The Royal Palace of Queen's Kingdom has a grand square chessboard-shaped security hall with 4 rooms arranged in 4 rows and 4 columns. Each queen represents a security officer who can monitor all rooms in the same row, column, or diagonal. [4] CLO4/ PLO-c

The kingdom wants to assign 4 security officers in such a way that no two officers can monitor each other, ensuring maximum privacy and non-overlapping watch zones. However, due to restricted royal zones, room (2,3) that means second row and third column cannot be assigned to any officer. This restricted room is marked as blocked (e.g., rooms under renovation or containing royal treasures).

Now, find one possible arrangement using an appropriate algorithm. (Visualize each step with detail).

b. A treasure hunter is carrying a small bag that can hold items mentioned in Table 4. They have found **four treasures**, each with a specific weight: [6] CLO4/ PLO-c

Table 4: List of treasures

Treasure	**Weight (kg)**
Gold Coin	3
Silver Ring	4

Diamond Necklace	5
Ruby Stone	2

The treasure hunter wants to select a subset of treasures such that the total weight is exactly 9 kg so the bag is perfectly balanced. Using the backtracking approach, Identify all possible subsets of treasures that sum exactly to 9 kg. Sub Set Sum.

QUESTION 5 [10 MARKS]

a. A national courier service in Bangladesh needs to plan a daily delivery route starting from its main hub in Dhaka, visiting regional centers (Chattogram, Sylhet, Rajshahi, Barishal, etc.) exactly once before returning to Dhaka. The company has a distance matrix satisfying the triangle inequality and requires a quick, near-optimal route update each day. Construct a near-optimal route with stepwise explanation using an efficient approximation algorithmic approach based on Figure 2. [5] CLO4/ PLO-c

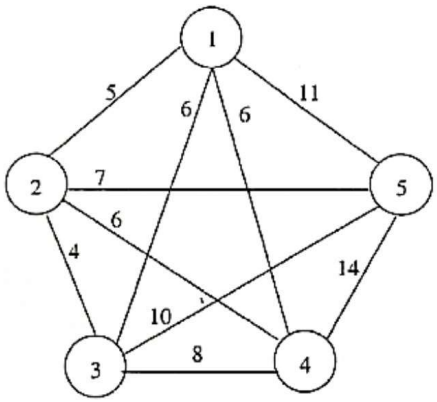


Figure 2. Graph of different connected districts

b. Mention how close to optimal (in terms of approximation ratio) the route can be of Question 5.(a), assuming all distances satisfy the triangle inequality. [2] CLO3/ PLO-b

c. Explain the difference among NP, NP hard and NP complete problems. [3] CLO1/ PLO-a

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
2nd Year 2nd Semester

Course Title: Probability and Statistics | Course Code: MTH 203
Time: 3 hours | Credit Hour: 3.00 | Full Marks: 50

There are **Five (5)** questions. Answer all the following questions. All questions are of equal value. Figures in the right margin indicate full marks.

QUESTION 1 [10 MARKS]

a. Determine the appropriate coefficient of skewness and kurtosis for the following frequency distribution: [5]

Age below (yrs)	25	30	35	40	45	50	55
No. of employees	8	20	40	65	80	92	100

b. Define standard deviation and coefficient of variation. The number of employees, average monthly wages per employee, and standard deviations of the two factories are given below: [5]

	Factory A	Factory B
Number of employees	100	150
Average monthly wages per employee (Tk)	3200	2800
Standard deviations (Tk)	625	729

i. Which factory A or B pays more monthly wages?
ii. Which factory shows greater variability in the distribution of wages?

QUESTION 2 [10 MARKS]

a. Find the mean deviation about the mean and the standard deviation from the data: [5]

Monthly Income	5–9	10–14	15–19	20–24	25–29	30–34
No. of family	15	30	55	17	10	3

Show that the standard deviation is greater than the mean deviation.

b. Define Karl Pearson's coefficient of correlation. Calculate Karl Pearson's coefficient of correlation from the following data on demand and price of a specific commodity. Represent the result graphically and comment on it. [5]

x :	10	12	21	4	7
y :	8	7	17	2	6

QUESTION 3 [10 MARKS]

a. Define mutually exclusive events and Bayes' theorem. In a bolt factory, machines A, B, and C manufacture, respectively 25%, 35%, and 40% of the total. If their output is 5, 4, and 2 percent, they are defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured [5]

by machine B?

b. In an industry, there is a 20% chance of suffering an occupational disease. Find the probability that out of 6 workers, (i) 3 workers will contract the disease, (ii) 4 workers will contract the disease, and (iii) 4 or more workers will contract the disease. [5]

QUESTION 4 [10 MARKS]

a. The probability function of a continuous random variable X is [5]

$$f(x)=\begin{cases}2(1-x), & 0<x<1\\ 0, & \text{elsewhere}\end{cases}$$

Find $E(X+1)$ and $V(X+1)$.

b. Define a joint probability mass function. A coin is drawn three times. If X denotes the number of heads in the last two tosses and Y denotes the number of tails tosses, find the joint probability distribution for X and Y. [5]

QUESTION 5 [10 MARKS]

a. A manufactured product has two defects per unit of product inspected. Find the probability of finding a product, [5]

i. No defect
ii. 3 defected
iii. 4 defected

b. Define normal and standard normal distribution. An electric firm manufactures a 100-watt light bulb which according to specifications written on the package, has a mean life of 900 hours with a standard deviation of 50 hours. At most, what percentage of the bulbs fail to last even 780 hours? Assume that the distribution is symmetric about the mean. [5]

OR

a. What are the degrees of freedom and level of significance? A random sample of size 16 values from a normal population showed a mean of 53 and a sum of squares of deviation from the mean equal to 150. Can this sample be regarded as taken from the population having 56 a mean? Obtain 95% and 99% confidence limits of the mean of the population. [5]

$$\gamma = 15, \quad \alpha = 0.05, \ t = 2.131$$

$$\alpha = 0.01, \ t = 2.947$$

b. A company pays its employees an average wage of \$5.25 per hour with a standard deviation of \$0.6. If the wage is normally distributed, what percentage of the employees receive wages between \$4.75 and \$5.69 per hour? [5]

Note: The necessary chart is attached

Table: Area under the normal curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998